

Revisiting Biomarker Panels under Wearable Constraints

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Abstract

This paper reports a systematic evaluation of patient-specific electrode arrays. Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol.

1. Introduction

Longer observation windows narrow the confidence interval but do not shift the central estimate. Calibration drift over the trial period remained below the threshold set out in the protocol. Neither of the alternative formulations improved on the simple additive model in our setting. Taken together, these findings argue for treating the two regimes separately in future analyses.

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Longer observation windows narrow the confidence interval but do not shift the central estimate. Calibration drift over the trial period remained below the threshold set out in the protocol. Sensitivity to the smoothing window proved modest across the range we examined. Taken together, these findings argue for treating the two regimes separately in future analyses. The reduced model retains most of the explanatory power at a fraction of the computational cost. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We release the processing scripts and intermediate artefacts to support independent verification. The measurements reported here were collected over twelve independent trials under controlled conditions. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting. Residual error grows approximately linearly with the number of coupled parameters.

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2. Related Work

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3. Materials and Methods

Replicate agreement was high, suggesting the protocol is robust to small handling differences. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Sensitivity to the smoothing window proved modest across the range we examined. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate. We release the processing scripts and intermediate artefacts to support independent verification. Neither of the alternative formulations improved on the simple additive model in our setting. Calibration drift over the trial period remained below the threshold set out in the protocol.

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4. Experimental Setup

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The measurements reported here were collected over twelve independent trials under controlled conditions. Calibration drift over the trial period remained below the threshold set out in the protocol. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

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5. Results

Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses. Calibration drift over the trial period remained below the threshold set out in the protocol. The reduced model retains most of the explanatory power at a fraction of the computational cost. Longer observation windows narrow the confidence interval but do not shift the central estimate. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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6. Discussion

Sensitivity to the smoothing window proved modest across the range we examined. Taken together, these findings argue for treating the two regimes separately in future analyses. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Neither of the alternative formulations improved on the simple additive model in our setting. Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost.

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7. Limitations

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8. Conclusion

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